

Two Extensors of Metrics

by

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Presented by K. BORSUK on November 13, 1979

Summary. By $\text{Metr}(Z)$ (resp. $\text{Metr}_c(Z)$) we denote the family of all compatible metrics (resp. complete compatible metrics) on a given metrizable space Z . Let A be a closed subset of a metrizable space X . In this note we construct two extensors of metrics $\Phi_1, \Phi_2 : \text{Metr}(A) \rightarrow \text{Metr}(X)$ satisfying the conditions

(i) $\Phi_i(d)|_A = d$ for $d \in \text{Metr}(A)$ and $i = 1, 2$.

(ii) If X is complete-metrizable, then

$$\Phi_i(\text{Metr}_c(A)) \subset \text{Metr}_c(X) \quad \text{for } i = 1, 2.$$

(iii) Φ_1 is a Lipschitz map.

(iv) If $d_1 \leq d_2$ then $\Phi_2(d_1) \leq \Phi_2(d_2)$ for $d_1, d_2 \in \text{Metr}(A)$.

(v) If $d_n(x, y) \rightarrow d_0(x, y)$ for $x, y \in A$, then

$$\Phi_2(d_n)(x, y) \rightarrow \Phi_2(d_0)(x, y) \quad \text{for } x, y \in X.$$

In 1930 Hausdorff [4] proved that if A is a closed subset of a metrizable space X then any compatible metric d on A can be extended to a compatible metric $\tilde{d}$ on X . Since then the problem of extending metrics has been investigated by several authors [2, 8, 6, 5]. Bacon [1] observed that if X is complete-metrizable and d is complete then $\tilde{d}$ can be taken to be complete. The aim of this note is to construct two "canonical" extensors of metrics preserving, e.g. the inequalities of metrics, their point-wise convergence and other properties. The methods used extend those of [7].

Given a metrizable space X . By $\text{Metr}(X)$ (resp. $\text{Metr}_b(X)$, $\text{Metr}_{tb}(X)$, $\text{Metr}_c(X)$, $\text{Pm}(X)$) we denote the set of all compatible metrics (resp. bounded compatible metrics, totally bounded compatible metrics, complete compatible metrics, continuous pseudometrics) on X .

If $d_1, d_2 \in \text{Pm}(X)$ then

$$\|d_1 - d_2\| = \sup\{|d_1(x, y) - d_2(x, y)| : x, y \in X\}$$

defines a metric (which may take the value $+\infty$) on $\text{Pm}(X)$, moreover $\|d\|$ is finite for $d \in \text{Metr}_b(X)$ and $\|d_1 - d_2\|$ may be finite for non-bounded pseudometrics $d_1, d_2 \in \text{Pm}(X)$.

A map $\Phi : \text{Pm}(X) \rightarrow \text{Pm}(Y)$ is Lipschitz iff there exists $C > 0$ such that

$$\|\Phi(d_1) - \Phi(d_2)\| \leq C\|d_1 - d_2\| \quad \text{for } d_1, d_2 \in \text{Pm}(X).$$

Let X be a metrizable space and A be a closed subset of X . In this note we prove the following two theorems

THEOREM 1. *There exists an extensor $\Phi : \text{Pm}(A) \rightarrow \text{Pm}(X)$ satisfying the conditions*

- (1-i) $\Phi(d)|_A = d$ for $d \in \text{Pm}(A)$.
- (1-ii) $\Phi(\text{Metr}(A)) \subset \text{Metr}(X)$.
- (1-iii) Φ is a Lipschitz map.
- (1-iv) $\Phi(\text{Metr}_b(A)) \subset \text{Metr}_b(X)$.
- (1-v) If X is separable then $\Phi(\text{Metr}_{tb}(A)) = \text{Metr}_{tb}(X)$.
- (1-vi) If X is complete-metrizable then $\Phi(\text{Metr}_c(A)) = \text{Metr}_c(X)$.

THEOREM 2. *There exists an extensor $\Phi : \text{Pm}(A) \rightarrow \text{Pm}(X)$ satisfying the conditions*

- (2-i) $\Phi(d)|_A = d$ for $d \in \text{Pm}(A)$.
- (2-ii) $\Phi(\text{Metr}(A)) \subset \text{Metr}(X)$.
- (2-iii) If $d_1 \leq d_2$ then $\Phi(d_1) \leq \Phi(d_2)$ for $d_1, d_2 \in \text{Pm}(A)$.
- (2-iv) $\Phi(\text{Metr}_b(A)) \subset \text{Metr}_b(X)$.
- (2-v) If X is separable then $\Phi(\text{Metr}_{tb}(A)) \subset \text{Metr}_{tb}(X)$.
- (2-vi) If X is complete-metrizable then $\Phi(\text{Metr}_c(A)) \subset \text{Metr}_c(X)$.
- (2-vii) If $d_n(x, y) \rightarrow d_0(x, y)$ for $x, y \in A$ then

$$\Phi(d_n)(x, y) \rightarrow \Phi(d_0)(x, y) \quad \text{for } x, y \in X.$$

1. Proof of Theorem 1. Consider the locally convex space $E = m(\text{Pm}(A), l^\infty(A))$ of all maps from $\text{Pm}(A)$ into $l^\infty(A)$ equipped with the topology of point-wise convergence. Fix a base point $x_0 \in A$ and define a map $g : A \rightarrow E$ by

$$(1-1) \quad g_x(d)(a) = d(x, a) - d(x_0, a) \quad \text{for } d \in \text{Pm}(A) \text{ and } a \in A.$$

Obviously g is continuous. By Dugundji's theorem [3] there exists a continuous map $\tilde{g} : X \rightarrow E$ such that $\tilde{g}|_A = g$ and $\tilde{g}(X) \subset \text{conv } g(A)$ (the convex hull of $g(A)$ in E).

Fix a compatible metric ρ on X which is complete if X is complete-metrizable and totally bounded if X is separable and define $\Phi_1, \Phi_2, \Phi : \text{Pm}(A) \rightarrow \text{Pm}(X)$ by

$$(1-2) \quad \Phi_1(d)(x, y) = \|g_x(d) - g_y(d)\|$$

$$(1-3) \quad \Phi_2(d)(x, y) = \sup\{|\theta(d)(x, z)\theta(d)(x, A) - \theta(d)(y, z)\theta(d)(y, A)| : z \in X\},$$

where

$$(1-4) \quad \theta(d) = \min\{1, \Phi_1(d) + \rho\}$$

$$(1-5) \quad \Phi(d) = \Phi_1(d) + \Phi_2(d).$$

Obviously Φ satisfies (1-i).

Proof of (1-ii). Since $\theta(\text{Pm}(A)) \subset \text{Metr}(X)$, it suffices to show that $\Phi(d)$ is equivalent to $\theta(d)$ for every $d \in \text{Metr}(A)$.

Suppose that $\theta(d)(x_n, x) \rightarrow 0$. By (1-4) and (1-3) we have $\Phi_i(d)(x_n, x) \rightarrow 0$ for $i = 1, 2$. Thus $\Phi(d)(x_n, x) \rightarrow 0$ by (1-5).

Conversely, assume to the contrary that for some sequence $(x_n) \subset X$ we have $\Phi(d)(x_n, x) \rightarrow 0$ and

$$(1-6) \quad \theta(d)(x_n, x) \geq \delta > 0 \quad \text{for every } n \in N.$$

Since by (1-3)

$$\theta(d)(x_n, x) \max\{\theta(d)(x, A), \theta(d)(x_n, A)\} \leq \Phi_2(d)(x_n, x) \leq \Phi(d)(x_n, x) \rightarrow 0$$

we infer that $x \in A$ and $\theta(d)(x_n, A) \rightarrow 0$.

Select a sequence of points $(x'_n) \subset A$ such that

$$(1-7) \quad \theta(d)(x_n, x'_n) \rightarrow 0$$

then $\Phi_1(d)(x_n, x'_n) \rightarrow 0$, by (1-4).

Since $\Phi(d)(x_n, x) \rightarrow 0$, from (1-5) we get $\Phi_1(d)(x_n, x) \rightarrow 0$ and

$$(1-8) \quad d(x'_n, x) = \Phi_1(d)(x'_n, x) \rightarrow 0.$$

Note that $x \in A$, $(x_n) \subset A$ and $d, \theta(d)|_A \in \text{Metr}(A)$, from (1-4) and (1-8) it follows that $\theta(d)(x'_n, x) \rightarrow 0$. Combining this with (1-7) we obtain $\theta(d)(x_n, x) \rightarrow 0$, which contradicts (1-6).

Proof of (1-iii). From (1-1) we get

$$(1-9) \quad \|g_x(d_1) - g_x(d_2)\| \leq 2\|d_1 - d_2\| \quad \text{for every } x \in A.$$

Since $\tilde{g}(X) \subset \text{conv } g(A)$, for every $x, y \in X$ we have

$$g_x = \sum_{i=1}^n \lambda_i g_{x_i}, \quad x_i \in A, \quad \lambda_i \geq 0 \text{ for } i = 1, \dots, n \text{ and } \sum_{i=1}^n \lambda_i = 1,$$

$$g_y = \sum_{j=1}^m \mu_j g_{y_j}, \quad y_j \in A, \quad \mu_j \geq 0 \text{ for } j = 1, \dots, m \text{ and } \sum_{j=1}^m \mu_j = 1.$$

Thus from (1-9) and (1-2) we obtain

$$\|\Phi_1(d_1) - \Phi_1(d_2)\| \leq 4\|d_1 - d_2\|.$$

On the other hand, from (1-3) and (1-4) we get

$$\|\Phi_2(d_1) - \Phi_2(d_2)\| \leq 4\|\theta(d_1) - \theta(d_2)\| \leq 4\|\Phi_1(d_1) - \Phi_1(d_2)\| \leq 16\|d_1 - d_2\|.$$

Thus by (1-5)

$$\|\Phi(d_1) - \Phi(d_2)\| \leq 20\|d_1 - d_2\| \quad \text{for } d_1, d_2 \in \text{Pm}(A)$$

(1-iv) follows from the relation $\tilde{g}(X) \subset \text{conv } g(A)$.

Proof of (1-v). Assume that ρ and d are totally bounded. Since

$$\|g_x(d) - g_y(d)\| = d(x, y)$$

the set

$$g(A)d = \{g(d) : a \in A\} \subset l^\infty(A)$$

is totally bounded. Thus $\tilde{g}(X)d \subset \text{conv } g(A)d$ is also totally bounded. Hence by (1-2) $\Phi_1(d)$ is totally bounded. Consequently by (1-4) $\theta(d)$ is totally bounded. Since $\Phi_2(d) \leq 2\theta(d)$ we infer that

$$\Phi(d) = \Phi_1(d) + \Phi_2(d)$$

is totally bounded.

Proof of (1-vi). By (1-4) if ρ is complete then $\theta(d)$ is also a complete metric for every $d \in \text{Pm}(A)$. Thus from the equivalence of $\Phi(d)$ and $\theta(d)$ it suffices to show that if $\Phi(d)(x_n, x_m) \rightarrow 0$ then $\theta(d)(x_n, x_m) \rightarrow 0$ for every $d \in \text{Metr}_c(A)$.

If it is not the case then there exist two subsequences $(a_n) \subset (x_n)$ and $(b_n) \subset (x_n)$ such that

$$(1-10) \quad \theta(d)(a_n, b_n) \geq \delta > 0 \quad \text{for every } n \in N.$$

Since by (1-3)

$$\theta(d)(a_n, b_n) \theta(d)(a_n, A) \leq \Phi_2(d)(a_n, b_n) \leq \Phi(d)(a_n, b_n) \rightarrow 0$$

we infer that there exists a sequence (y_n) of points in A such that

$$(1-11) \quad \theta(d)(a_n, y_n) \rightarrow 0.$$

Thus from (1-4) $\Phi_1(d)(a_n, y_n) \rightarrow 0$. Since $\Phi_1(d)|_A = d$ is complete and since

$$\begin{aligned} \Phi_1(d)(y_n, y_m) &\leq \Phi_1(d)(y_n, a_n) + \Phi_1(d)(a_n, a_m) + \Phi_1(d)(a_m, y_m) \\ &\leq \Phi_1(d)(y_n, a_n) + \Phi(d)(a_n, a_m) + \Phi_1(d)(a_m, y_m) \rightarrow 0 \end{aligned}$$

we infer that there exists a point $x_0 \in A$ such that

$$\Phi(d)(y_n, x_0) = \Phi_1(d)(y_n, x_0) = d(y_n, x_0) \rightarrow 0.$$

Thus $\theta(d)(y_n, x_0) \rightarrow 0$. Hence from (1-11) we get $\theta(d)(a_n, x_0) \rightarrow 0$. Consequently $\Phi(d)(a_n, x_0) \rightarrow 0$. Since (x_n) is a $\Phi(d)$ -Cauchy sequence and $(a_n) \subset (x_n)$ we infer that $\Phi(d)(x_n, x_0) \rightarrow 0$. Thus $\theta(d)(x_n, x_0) \rightarrow 0$. This contradicts (1-10) and concludes the proof of Theorem 1.

2. Proof of Theorem 2. For every $d \in \text{Pm}(A)$ let $\text{Lip}(A, d)$ denote the set of all Lipschitz maps $f : (A, d) \rightarrow R^1$ vanishing at $a_0 \in A$ equipped with the norm

$$\|f\|_d = \inf\{C : f \text{ is a } C\text{-Lipschitz map}\}.$$

Let $L(A)$ denote the linear space spanned formally by A . For every $d \in \text{Pm}(A)$ and $x = \sum_{i=1}^n \lambda_i a_i \in L(A)$, put

$$\|x\|_d = \sup \left\{ \left| \sum_{i=1}^n \lambda_i f(a_i) \right| : f \in \text{Lip}(A, d), \|f\|_d \leq 1 \right\}.$$

Then $L(A)$ becomes a locally convex space. By Dugundji's theorem [3] there exists a continuous map $\varphi : X \rightarrow L(A)$ such that $\varphi(a) = a$ for every $a \in A$ and $\varphi(X) \subset \text{conv}(A) \subset L(A)$.

Fix a compatible metric ρ on X which is complete if X is complete-metrizable and totally bounded if X is separable and define $\Phi_1, \Phi_2, \Phi : \text{Pm}(A) \rightarrow \text{Pm}(X)$ by

$$(2-1) \quad \Phi_1(d)(x, y) = \|\varphi(x) - \varphi(y)\|_d$$

$$(2-2) \quad \Phi_2(d)(x, y) = \sup\{|f(x) - f(y)| : f \in \text{Lip}(X, e), f|_A = 0 \text{ and } \|f\|_e \leq 1\},$$

where $e = \min\{1, \rho + \Phi_1(d)\}$

$$(2-3) \quad \Phi(d) = \Phi_1(d) + \Phi_2(d).$$

It is easy to see that Φ satisfies conditions (2-i) and (2-iii). The proofs of (2-ii), (2-iv), (2-v) and (2-vi) are similar to the proofs of (1-ii), (1-iv), (1-v) and (1-vi), respectively. Let us check (2-vii).

Assume that $d_n(x, y) \rightarrow d_0(x, y)$ for $x, y \in A$. We have to show that $\Phi_i(d_n)(x, y) \rightarrow \Phi_i(d_0)(x, y)$ for $x, y \in X$ and $i = 1, 2$.

Take $x_0, y_0 \in X$. Then we have

$$\varphi(x_0) - \varphi(y_0) = \sum_{i=1}^k \lambda_i a_i,$$

where $a_1, a_2, \dots, a_k \in A$.

The case $i = 1$. For every $\varepsilon > 0$, take $p > \Phi_1(d_0)(x_0, y_0)$ and let $n_0 \in N$ be such that

$$(2-4) \quad \left(1 - \frac{\varepsilon}{p}\right) d_0(a, b) \leq d_n(a, b) \leq \left(1 + \frac{\varepsilon}{p}\right) d_0(a, b)$$

for every $n \geq n_0$ and $a, b \in A_0 = \{a_1, \dots, a_k\}$.

From (2-4) we get for $n \geq n_0$

$$\begin{aligned} \Phi_1(d_n)(x_0, y_0) &= \sup \left\{ \left| \sum_{i=1}^k \lambda_i f(a_i) \right| : f \in \text{Lip}(A, d_n) : \|f\|_{d_n} \leq 1 \right\} \\ &= \sup \left\{ \left| \sum_{i=1}^k \lambda_i f(a_i) \right| : f \in \text{Lip}(A_0, d_n), \|f\|_{d_n} \leq 1 \right\} \\ &\leq \sup \left\{ \left| \sum_{i=1}^k \lambda_i f(a_i) \right| : f \in \text{Lip}(A_0, d_0), \|f\|_{d_0} \leq 1 + \frac{\varepsilon}{p} \right\} \\ &= \left(1 + \frac{\varepsilon}{p}\right) \sup \left\{ \left| \sum_{i=1}^k \lambda_i f(a_i) \right| : f \in \text{Lip}(A_0, d_0), \|f\|_{d_0} \leq 1 \right\} \\ &= \left(1 + \frac{\varepsilon}{p}\right) \sup \left\{ \left| \sum_{i=1}^k \lambda_i f(a_i) \right| : f \in \text{Lip}(A, d_0), \|f\|_{d_0} \leq 1 \right\} \\ &= \left(1 + \frac{\varepsilon}{p}\right) \Phi_1(d_0)(x_0, y_0) < \Phi_1(d_0)(x_0, y_0) + \varepsilon. \end{aligned}$$

Similarly,

$$\Phi_1(d_n)(x_0, y_0) > \Phi_1(d_0)(x_0, y_0) - \varepsilon \quad \text{for every } n \geq n_0.$$

The case $i = 2$. Let $e_n(x, y) = \rho(x, y) + \Phi_1(d_n)(x, y)$. By the case $i = 1$ we get

$$e_n(x, y) \rightarrow e_0(x, y) \quad \text{for every } x, y \in X.$$

For every $\varepsilon > 0$, take an $n_0 \in \mathbb{N}$ such that

$$(2-5) \quad \left(1 - \frac{\varepsilon}{p}\right) e_0(x, y) \leq e_n(x, y) \leq \left(1 + \frac{\varepsilon}{p}\right) e_0(x, y) \quad \text{for every } n \geq n_0,$$

where $p > \Phi_2(d_0)(x_0, y_0)$.

Thus for every $n \geq n_0$ we have

$$\begin{aligned} \Phi_2(d_n)(x_0, y_0) &= \sup\{|f(x_0) - f(y_0)| : f \in \text{Lip}(X, e_n), f|_A = 0 \text{ and } \|f\|_{e_n} \leq 1\} \\ &\leq \sup\left\{|f(x_0) - f(y_0)| : f \in \text{Lip}(X, e_0), f|_A = 0 \text{ and } \|f\|_{e_0} \leq 1 + \frac{\varepsilon}{p}\right\} \\ &= \left(1 + \frac{\varepsilon}{p}\right) \sup\{|f(x_0) - f(y_0)| : f \in \text{Lip}(X, e_0), f|_A = 0 \text{ and } \|f\|_{e_0} \leq 1\} \\ &= \left(1 + \frac{\varepsilon}{p}\right) \Phi_2(d_0)(x_0, y_0) < \Phi_2(d_0)(x_0, y_0) + \varepsilon. \end{aligned}$$

Similarly,

$$\Phi_2(d_n)(x_0, y_0) > \Phi_2(d_0)(x_0, y_0) - \varepsilon \quad \text{for every } n \geq n_0.$$

This completes the proof of Theorem 2.

The authors wish to express their deep gratitude to H. Toruńczyk for his kind interest during the preparation of this note.

Their thanks are due also to J. Luukkainen for many valuable remarks.

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Нгуен Ван Khue, Нгуен То Ну, О двух экстенсорах метрик.

Через $\text{Metr}(Z)$ (соответственно, $\text{Metr}_c(Z)$) мы обозначаем семейство всех компактизуемы метрик (соответственно, вполне компактизуемых метрик) на данном метризуемом пространстве Z . Пусть A замкнутое подмножество метризуемого пространства X . В данной работе строятся два экстенсора метрик $\Phi_1, \Phi : \text{Metr}(A) \rightarrow \text{Metr}(X)$, удовлетворяющих условиям:

- (i) $\Phi_i(d)|_A = d$, где $d \in \text{Metr}(A)$, $i = 1, 2$;
- (ii) если X вполне метризуемо, то $\Phi_i(\text{Metr}_c(A)) \subset \text{Metr}_c(X)$ $i = 1, 2$;
- (iii) Φ_1 есть Липшицово отображение;
- (iv) если $d_1 \leq d$, то $\Phi(d_1) \leq \Phi(d)$ где $d_1, d \in \text{Metr}(A)$;
- (v) если $d_n(x, y)$, где $x, y \in A$, то $\Phi(d_n)(x, y) \rightarrow \Phi(d)(x, y)$, где $x, y \in X$.